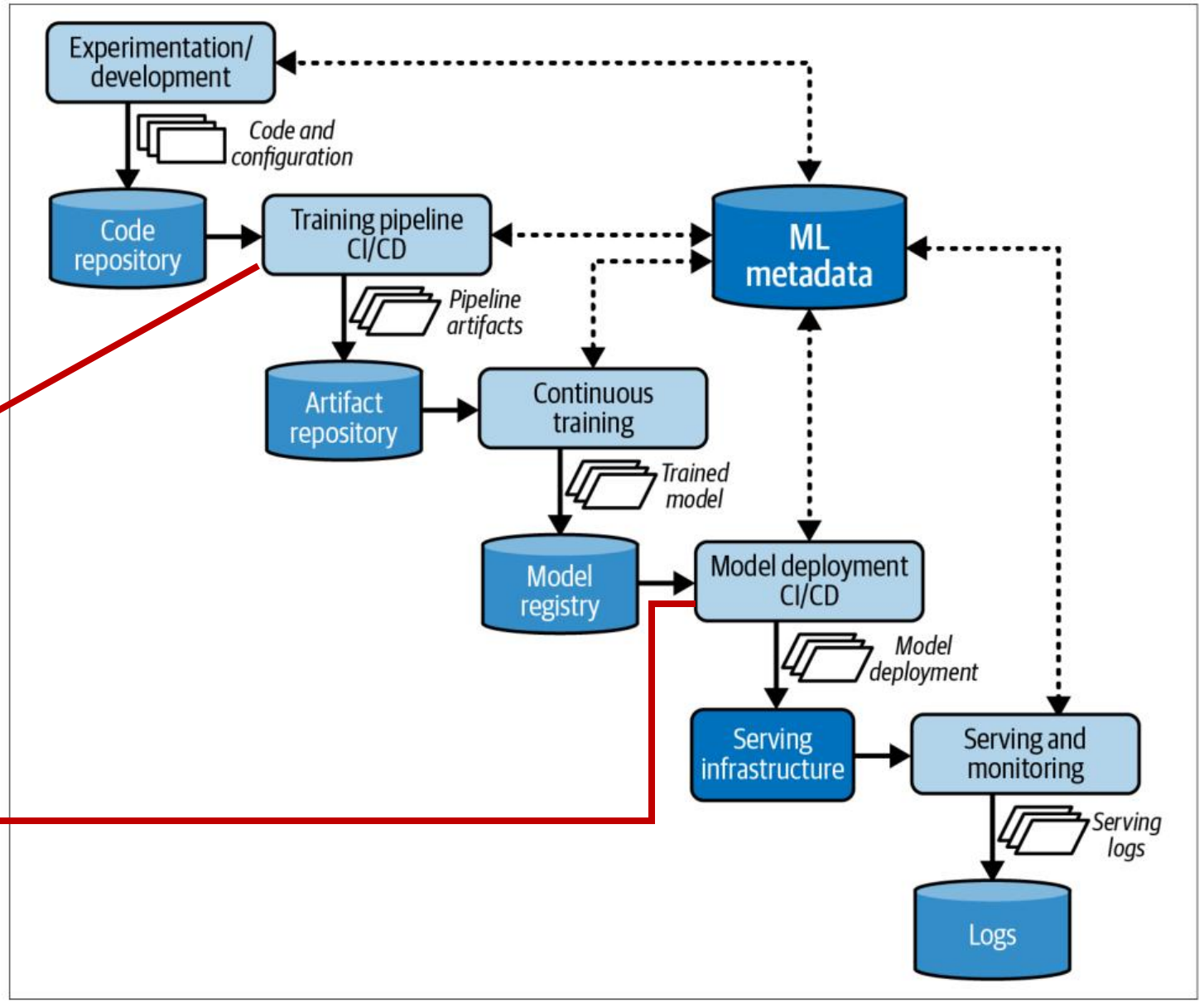
CI/CD with Jenkins in MLOps

Mohammad Fozouni (Ph.D.)

MLSECOPS IN ACTION COURSE




Experimentation/ development
Code and configuration
Code repository
Training pipeline CI/CD
ML metadata
Pipeline artifacts
Artifact repository
Continuous training
Trained model
Model registry
Model deployment CI/CD
Model deployment
Serving infrastructure
Serving and monitoring
Serving logs
Logs



Jenkins is an open-source **automation server** that helps developers **build, test, and deploy** code automatically through continuous integration and continuous deployment (CI/CD) pipelines.

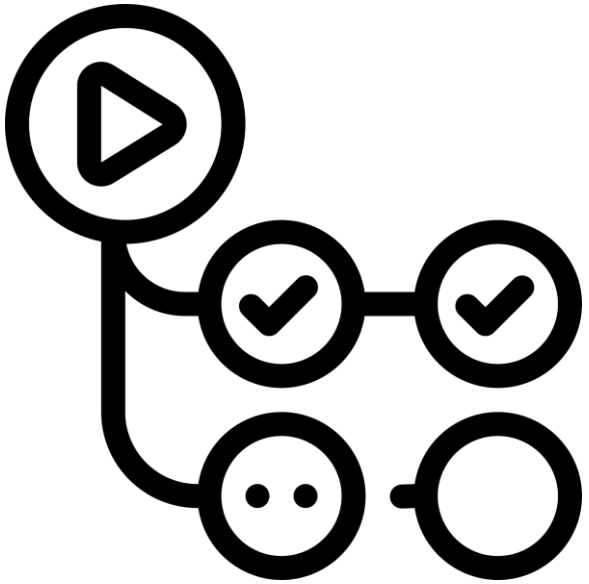
It monitors code repositories for changes and triggers automated workflows to compile, test, and release software, making the development process faster and more reliable.



Jenkins is an open-source **automation server** that helps developers **build, test, and deploy** code automatically through continuous integration and continuous deployment (CI/CD) pipelines.

It monitors code repositories for changes and triggers automated workflows to compile, test, and release software, making the development process faster and more reliable.

Similar to GitHub Actions 👍

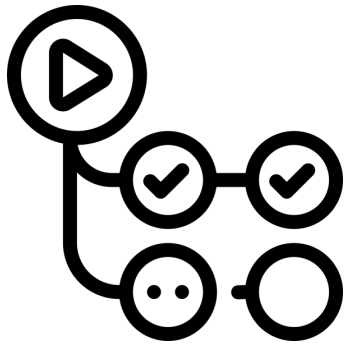




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It monitors code repositories for changes and triggers automated workflows to compile, test, and release software, making the development process faster and more reliable.

Similar to GitHub Actions 👍



BUT we have more CONTROLS

What we want to do in this session 🤔

Server: 12.54.32.56

http://12.54.32.56:5000





http://12.54.32.56:8080



Server

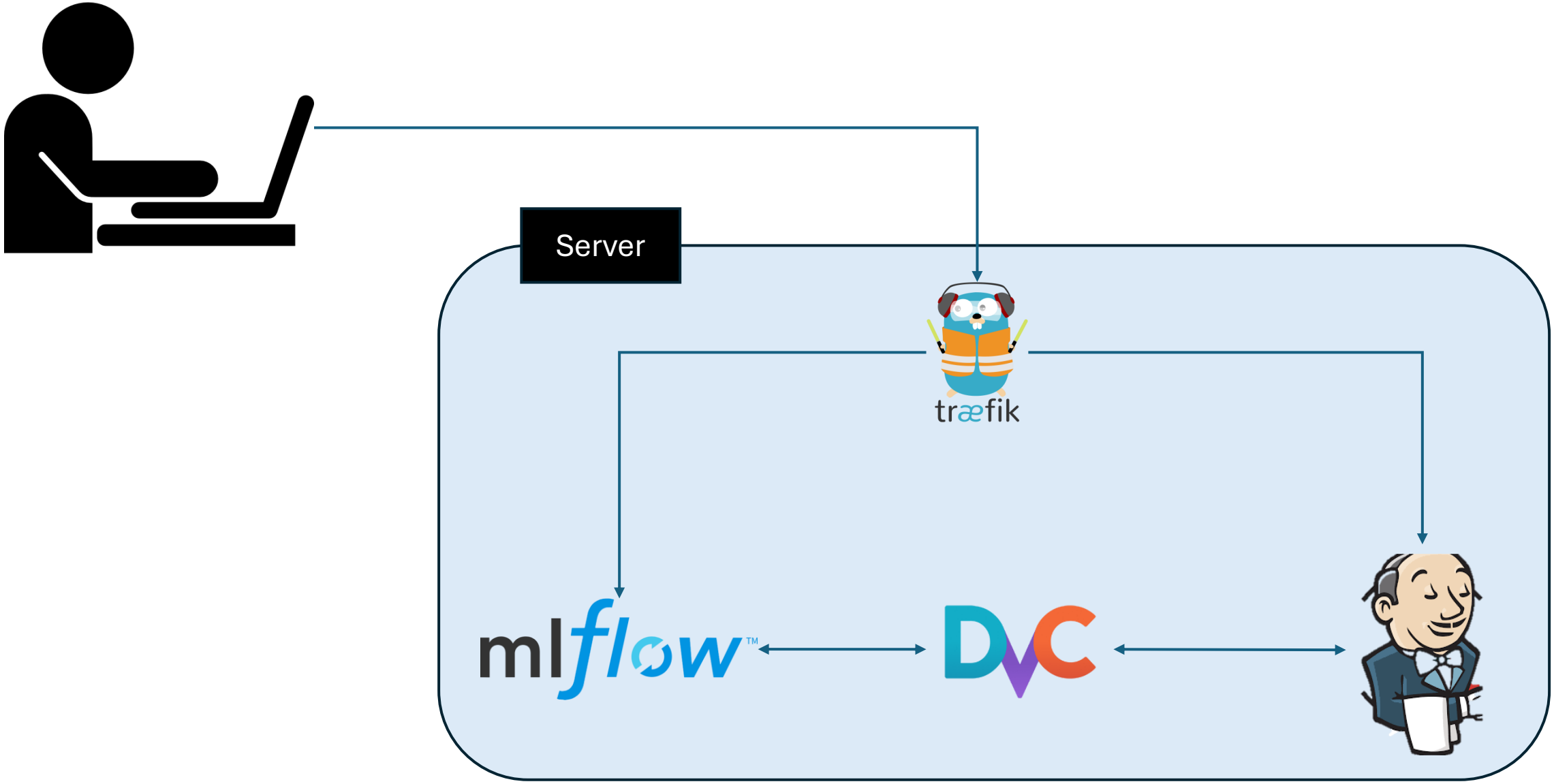


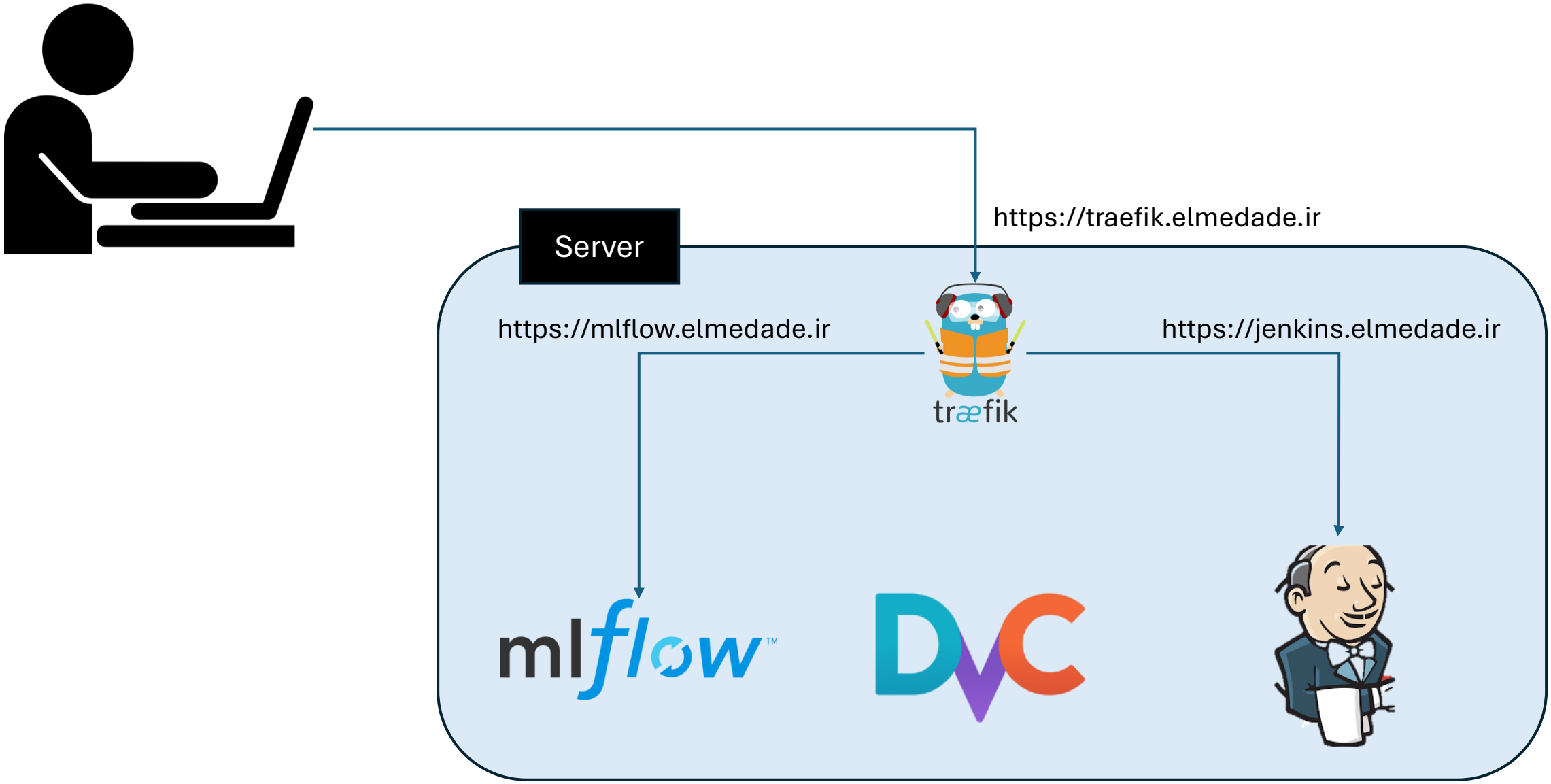
traefik

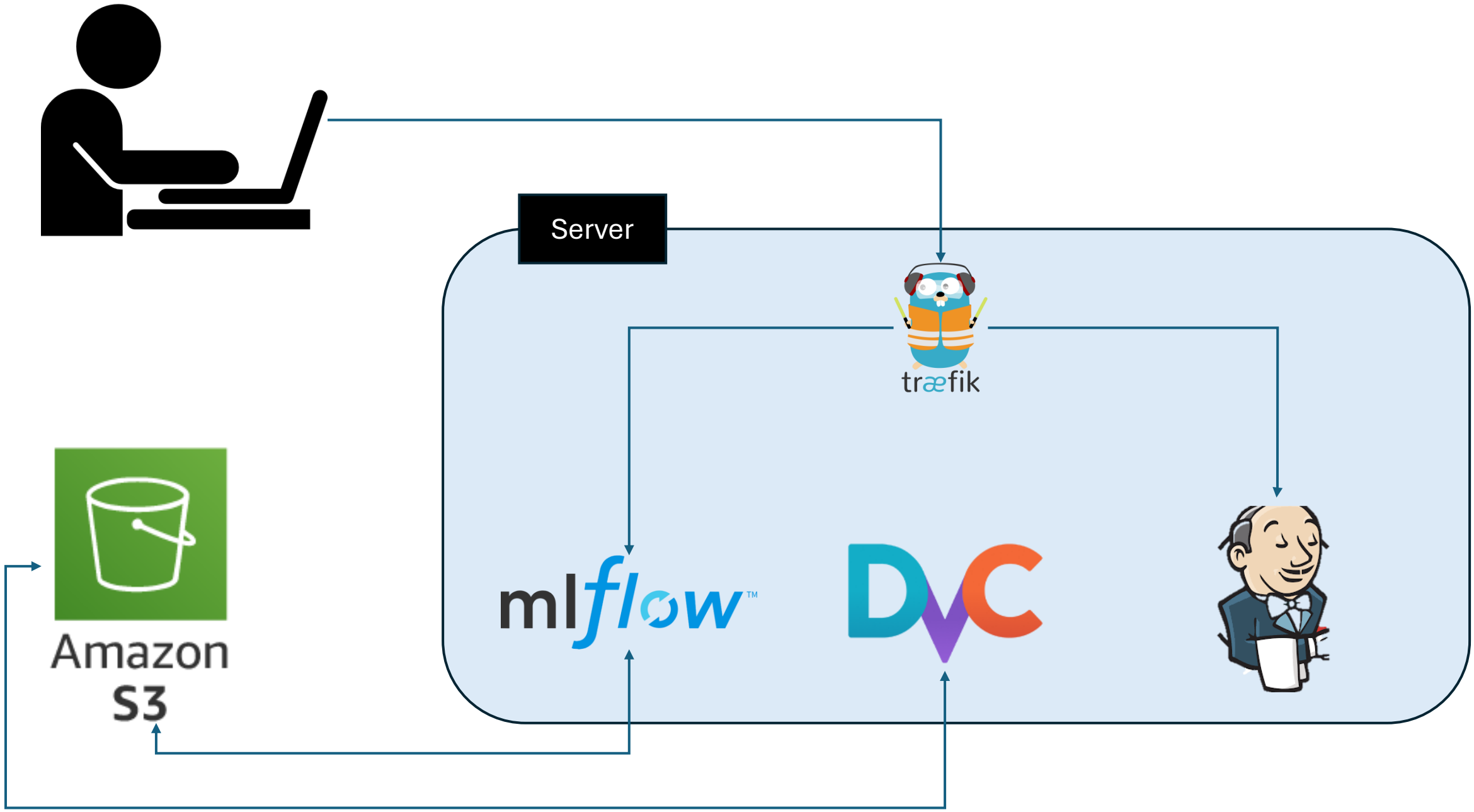
mlflow

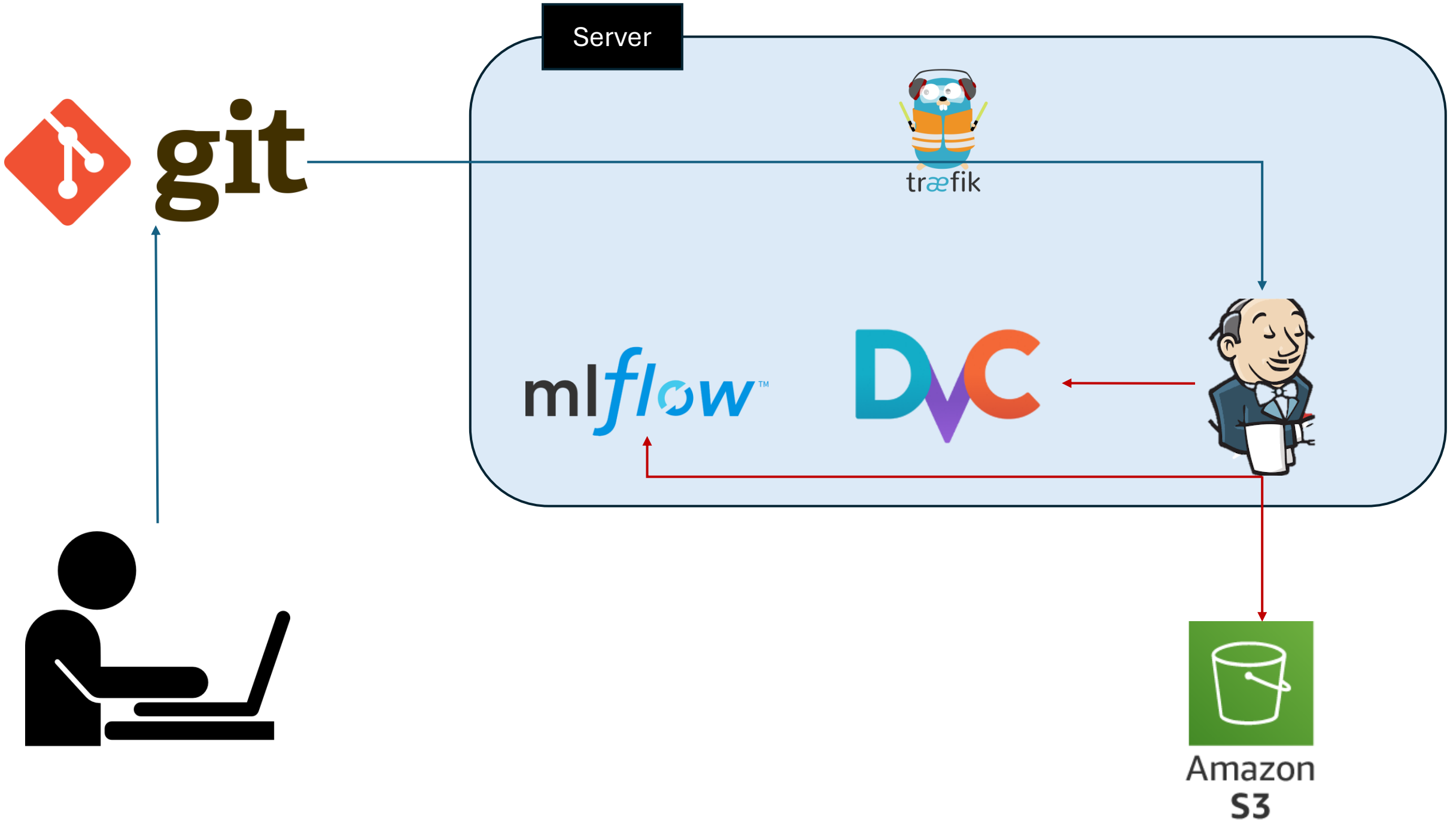
DVC




Server
træfik
mlflow™
DVC


Server
https://traefik.elmedade.ir
https://mlflow.elmedade.ir
https://jenkins.elmedade.ir
træfik
mlflow
DVC


Server
træfik
mlflow
DVC
Amazon
S3


Server
git
træfik
mlflow
DVC
Amazon
S3

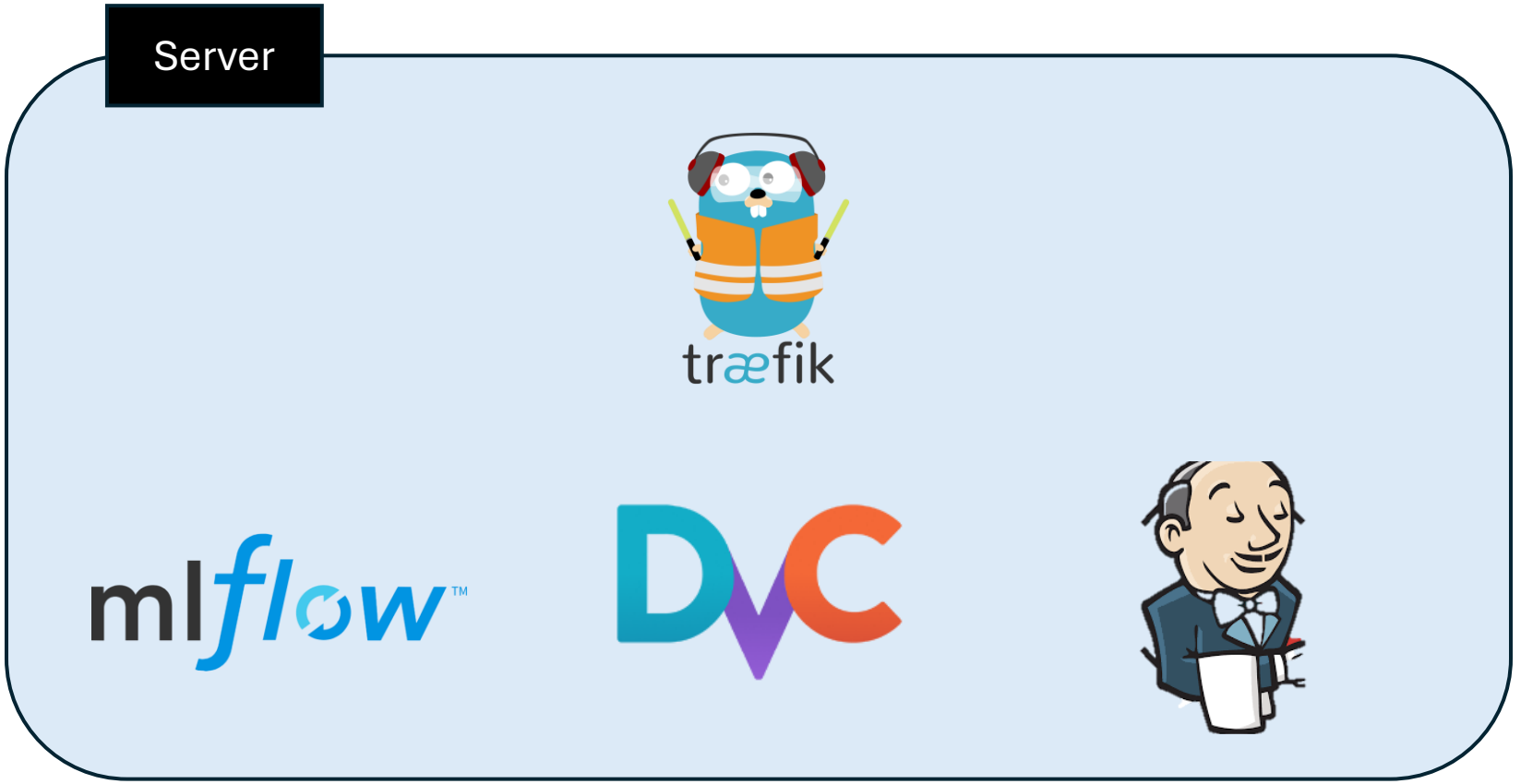
Our architecture is Production-Ready?

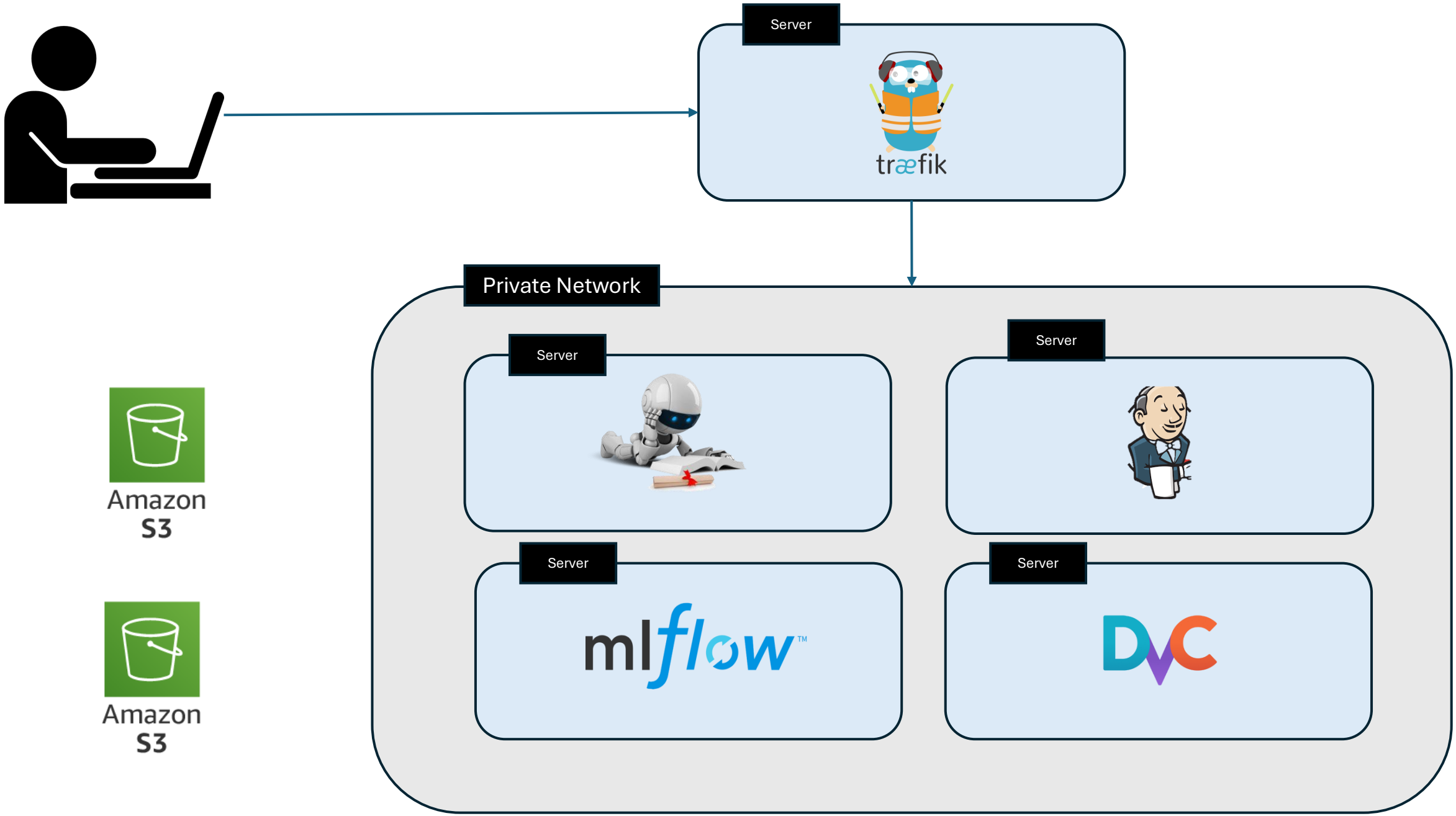
Somehow 😎

Because here we have SPOF.

If our server goes down, we will also






Server
træfik
Private Network
Server
Server
Server
Server
mlflow
DVC
Amazon
S3
Amazon
S3

How we set DNS and A records of our Domain

If you have a running website

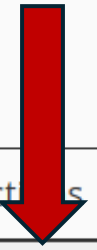
Zone Editor

Domains

DNS converts domain names into computer-readable IP addresses. Use this feature to manage DNS zones. For more information, read the documentation.

Domains

Filter by domain



Domain	Actions
m-fozouni.ir	+ A Record + CNAME Record + MX Record DNSSEC Manage

Zone Editor

Domains

DNS converts domain names into

more information, read the documentation

Domains

Filter by domain

Domain	Actions
m-fozouni.ir	+ A Record + CNAME Record + MX Record DNSSEC Manage

Add an A Record for “m-fozouni.ir”

Name

example.m-fozouni.ir.

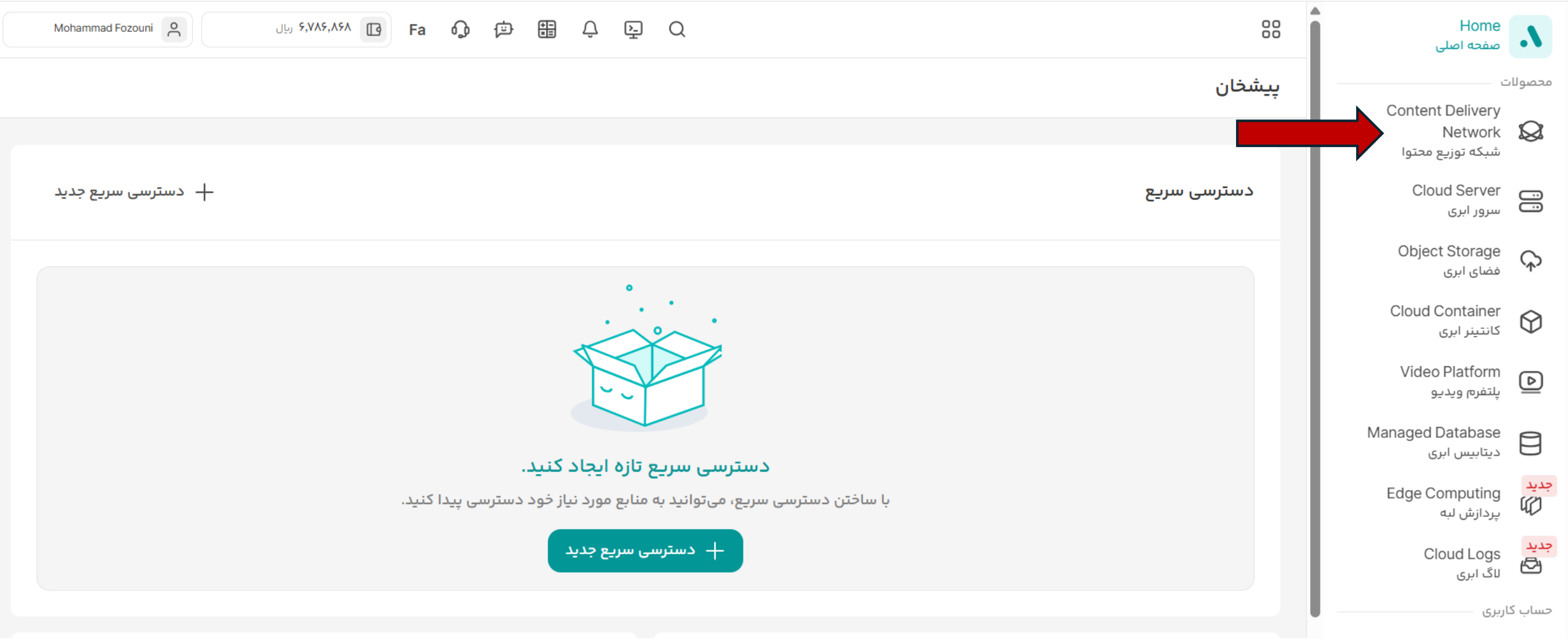
Address

203.0.113.11

Add An A Record

Cancel

If you have an empty domain



شبکه توزیع محتوا
Content Delivery Network

لیست دامنه‌ها

دامنه‌های من
انتقال دامنه
لیست‌ها
مانیتورینگ متریک‌های دامنه
خرید گواهی‌نامه‌ی SSL
گزارش‌ها
لاگ تغییرات پنل

۱ ورود دامنه — ۲ انتخاب پلن — ۳ رکورد های DNS — ۴ وضعیت دامنه



برای شروع نیاز به یک دامنه دارید
دامنه خود را بدون www وارد کنید

example.com

ادامه، به معنای پذیرش شرایط و مقررات استفاده از آروان‌کلاد است.

قبول قوانین و ادامه

انصراف

شما پیام جدید دارید.
برو به مرکز پیام‌ها

دامنه‌های من

- برای پرداخت درخواست‌ها از قبیل ثبت و تمدید از صفحه درخواست‌های من اقدام کنید.
- مهلت پرداخت دامنهٔ ثبت شده تا قبل از تاریخ ذکر شده در ستون «زمان انقضاء» می‌باشد.
- اگر درخواستی توسط مرکز ثبت دامنه تایید نشود، وجه آن به حساب کاربری پرداخت کننده بازگردانده خواهد شد.

نام دامنه:	◉ شروع با ○ شامل		جستجو »
شناسهٔ رابط:	هرکدام		

در حساب خود **2 دامنه** دارید.

فهرست تمامی دامنه‌ها

تعداد کل: **2** سطر — صفحه: **1** از **1** — برو به صفحهٔ: **1** — نمایش حداکثر **20** / 50 / 100 سطر

انتخاب	دامنه	صاحب‌امتیاز	وضعیت	آخرین تغییر	زمان انقضاء	دی‌ان‌اس (DNS)
☐	elmedade.ir	Mohammad Fozouni (mf10460-irnic)	ثبت شده	2025-08- 1	2026-05-22	NS h.ns.arvancdn.ir



Let’s GOOOOO
For coding 😎